



Walmart Sales Performance Analysis

2019–2023 End-to-End Data Analytics Project

A comprehensive analysis of Walmart's sales data spanning five years, transforming raw transactional data into actionable business insights using Python, SQL, and Tableau for data-driven decision making.



Business Objective & Questions

Objective

Analyze Walmart's sales data to understand **revenue drivers**, **customer behavior**, and **operational patterns** across all branches and time periods.

Key Business Questions

- Which branches contribute most to revenue?
- How has revenue changed over time?
- When do customers shop the most?
- Which payment methods dominate?
- Where are performance gaps and improvement opportunities?

Dataset Overview

9,969

Transactions

Total sales records analyzed

\$1.21M

Total Revenue

Five-year sales volume

\$121.35

Average Basket

Per transaction value

5.83

Customer Rating

Out of 10 points

Time Period: 2019–2023 | **Data Sources:** Branch, City, Product Category, Date & Time, Quantity, Payment Method, Customer Rating

Data Preparation & Tools

01

Data Collection

Dataset sourced from **Kaggle** and imported into VS Code for initial analysis and quality assessment

02

Data Cleaning (Python – Pandas)

Fixed date parsing issues, separated date & time components, standardized numeric columns, and created derived fields including hour, day, month, and basket value calculations

03

Analysis & Visualization

Conducted SQL analysis in **PostgreSQL** for querying and aggregation, built interactive dashboard in **Tableau** for stakeholder presentation



Overall Performance KPIs



Key Performance Indicators

- **\$1.21M** total revenue from **9,969** transactions
- Customers spend **\$121.35** per transaction on average
- Average rating of **5.83/10**, indicating moderate satisfaction

Key Insight

High average spend suggests strong revenue per customer, but **customer experience improvement** could unlock further growth potential.



Branch & Revenue Insights

Branch Performance

Top-performing branches generate **~15–20% more revenue** than the lowest-performing branches, revealing significant performance variance across locations.

Revenue Concentration

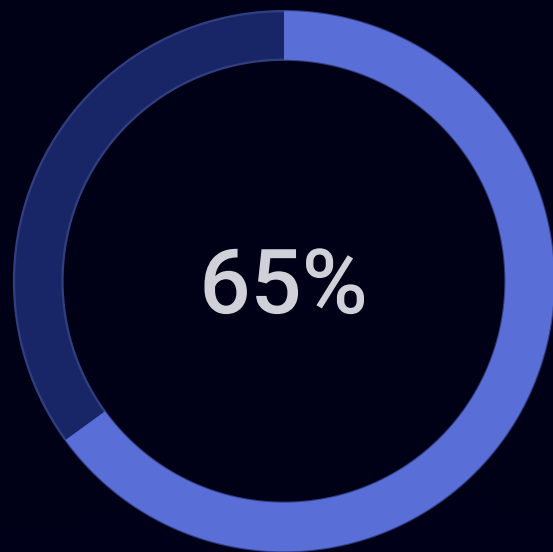
Revenue is **concentrated among a small number of branches**, indicating uneven performance distribution and untapped potential in underperforming locations.

Seasonal Patterns

Monthly revenue peaks exceed **\$60K** during year-end periods, with sharp declines followed by recovery indicating **seasonality and demand volatility**.

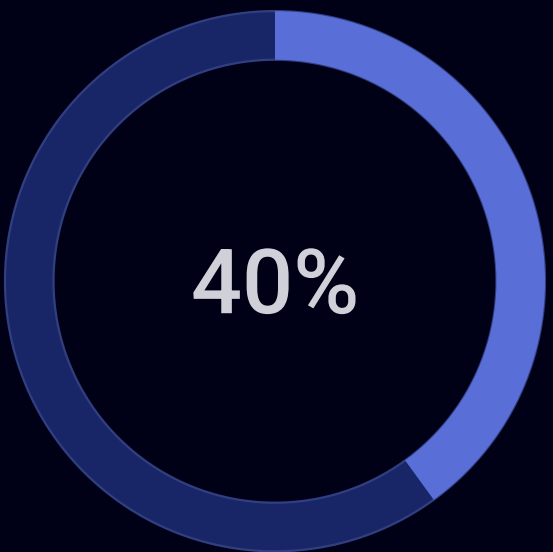
Customer Behavior Patterns

Time-Based Insights



Afternoon & Evening

Transactions occurring after midday



Peak Window

3 PM – 7 PM daily transactions

Day of Week Insights

Busiest Days

Tuesday and Thursday each contribute ~14–15% of weekly transactions, representing peak shopping activity

Slowest Day

Monday sees ~13% fewer transactions than peak weekdays, indicating lower customer engagement at week start

Payment & Category Insights

Payment Method Usage



Over 80% of transactions are digital (credit cards + e-wallets), reflecting strong adoption of electronic payment methods

Category Insights

High-revenue categories do not always align with high-profit categories, indicating opportunity to optimize **pricing and promotion strategies** for improved margin performance.

Key Recommendations



Operational Optimization

Increase staffing during **3–7 PM** peak window, which drives **40%+ of daily sales**. Use peak weekday insights (Tuesday/Thursday) to optimize inventory replenishment schedules.



Branch Strategy

Focus improvement efforts on branches generating **15–20% lower revenue**. Replicate best practices from top-performing branches to drive consistent performance across all locations.



Digital Payment Promotion

Promote digital payments (already **80%+ adoption**) via targeted incentives for remaining cash users. Continue building on strong digital payment foundation.



Customer Experience

Improve service quality to raise ratings by **1–1.5 points**, potentially boosting repeat purchases and long-term customer value through enhanced satisfaction.